Yiksan Yu

RBG Hall, Cornell, Ithaca, NY
14850
(646) 359-1298 yy2375@cornell.edu

EDUCATION

Cornell University, College of Engineering, Ithaca, NY ***Expected B.S. May 2028***
Bachelor of Science, Civil Engineering **GPA: 3.91**

Bronx High School of Science, Bronx, NY ***Diploma received June 2025***
Honors: AP scholars with distinction, National First-Generation recognition award **GPA: 4.00**

STEM EXPERIENCE

NYC Utilidor research, Cornell University, Ithaca, NY ***Sep 2025 – Present***

- Model underground infrastructure resilience by synthesizing New York City geological data to simulate structural responses under varied elastic and elastoplastic loading conditions.
- Optimize simulation workflows using the MOOSE framework and Gmsh to evaluate the structural integrity of tunnel systems, improving the accuracy of predictive models for Manhattan schist.

Cornell Seismic Design Project team, Cornell University, Ithaca, NY ***Oct 2025 - Present***

- Optimize structural frames in SAP2000 and AutoCAD to minimize displacement and resonance, directly increasing design precision for simulated high-magnitude seismic events.
- Engineer high-performance damping systems in Autodesk Fusion 360 to mitigate lateral load response and maximize energy dissipation during seismic loading.
- Validate structural integrity through rigorous shake-table testing, ensuring 100% adherence to competition safety requirements and load-bearing standards.
- Refine model reliability by correlating theoretical damping ratios with physical results, narrowing the margin of error between predicted and actual structural performance.

Johns Hopkins Summer institute, Baltimore, MD, *Summer lab student* ***July – Sep 2025***

- Constructed and tested bridge prototypes to maximize strength-to-material efficiency, applying structural mechanics principles and iterative design
- Significantly enhanced physical stability of various kinds of structures through calculation of the risk and safety of model infrastructure prototypes

LEADERSHIP EXPERIENCE

Independent SAT Tutoring Business, Online ***Feb – July 2025***

- Supervised a team of tutors, coordinating schedules and resolving conflicts to ensure consistent student progress.
- Developed media strategies that secured international clients from Japan, China, San Francisco, and New York
- Designed personalized learning plans and managed business operations, improved student performance and expanded client base by 300% within six months

COMMUNICATION

Bronx Science Varsity Policy Debate ***Sep 2021 – May 2024***

- Competed nationally for 3 years, honed research, public speaking, and teamwork skills
- Developed advanced research and technical communication skills applicable to design reviews and collaborative engineering projects
- Awards: National Forensic League *Degree of Distinction*; American Debate League – Top Team Award

SPECIALIZED SKILLS

Technical: GMSH, MOOSE(C++), AutoCAD, Prototyping (3D printing), Python, Fusion360, Sap2000
Languages: English (fluent), Chinese (fluent), Spanish (conversational)

VOLUNTEERING EXPERIENCE

Asian American Cohesion ***June 2022 – Sep 2022***

- Coordinated community service events and fostered cultural discussions to promote inclusivity and engagement.